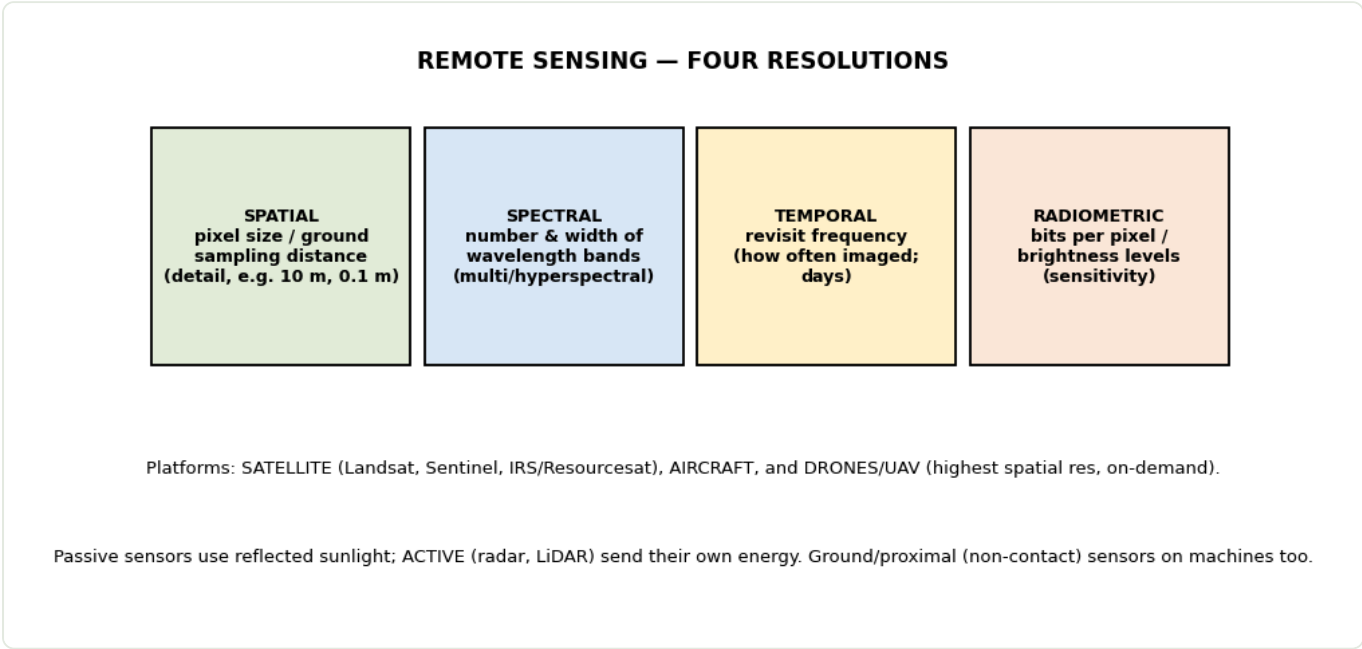
UNIT 10 · TOPIC 7

Data Analysis — Resolution and Surface Analysis

Syllabus topic: Data analysis concepts: resolution and surface analysis

The four resolutions of imagery



Remote sensing — the four resolutions

Resolution	Meaning	Example
SPATIAL	Pixel size / ground-sampling distance — the detail	Drones: centimetres; satellites: metres to tens of metres
SPECTRAL	Number and width of wavelength bands	Panchromatic vs multispectral vs hyperspectral
TEMPORAL	Revisit frequency — how often the same place is imaged	Days
RADIOMETRIC	Bits per pixel — number of brightness levels (sensitivity)	8-bit = 256 levels

From raw imagery to information

1. Geometric and atmospheric correction.
2. Classification.
3. Index computation (NDVI etc. — Topic 6).
4. Interpolation of point data into surfaces.
5. **SURFACE ANALYSIS** — derive slope, aspect and flow from elevation data.

Machine learning is increasingly used at each step (Unit 2, Topic 10).

Practice MCQs

Q45. Pixel size / ground sampling distance defines the ___ resolution:

(a) Spectral (b) Spatial
(c) Temporal (d) Radiometric

Q46. The number and width of wavelength bands defines the ___ resolution:

(a) Spatial (b) Spectral
(c) Temporal (d) Radiometric

Q47. How often a satellite images the same area is its ___ resolution:

(a) Spatial (b) Spectral
(c) Temporal (d) Radiometric

Q48. The number of brightness levels (bits/pixel) is the ___ resolution:

(a) Spatial (b) Spectral
(c) Temporal (d) Radiometric

Q70. Surface analysis in GIS derives, from elevation data:

(a) Slope, aspect and flow (b) NDVI
(c) Grain moisture (d) Octane number

Answers

Q45 — (b) Spatial — *Spatial = pixel size.*

Q46 — (b) Spectral — *Spectral = bands.*

Q47 — (c) Temporal — *Temporal = revisit.*

Q48 — (d) Radiometric — *Radiometric = brightness levels.*

Q70 — (a) Slope, aspect and flow

Sources: Srinivasan — Handbook of Precision Agriculture; ICAR/NIIT PA modules; Lillesand & Kiefer — Remote Sensing and Image Interpretation; ISO 11783 (ISOBUS); remote-sensing and NDVI references (as given in the source notes, July 2026); NPTEL precision-agriculture lectures.